Rich Kaplan–Meier Plot:
A Practical Extension of the Kaplan–Meier Curve

EPS Corporation
Yutaka Morioka

NONPARAMETRIC ESTIMATION FROM INCOMPLETE OBSERVATIONS*

E. L. KAPLAN
University of California Radiation Laboratory
AND
PAUL MEIER
University of Chicago

In lifetesting, medical follow-up, and other fields the observation of the time of occurrence of the event of interest (called a *death*) may be prevented for some of the items of the sample by the previous occurrence of some other event (called a *loss*). Losses may be either accidental or controlled, the latter resulting from a decision to terminate certain observations. In either case it is usually assumed in this paper that the lifetime (age at death) is independent of the potential loss time; in practice this assumption deserves careful scrutiny. Despite the resulting incompleteness of the data, it is desired to estimate the proportion $P(t)$ of items in the population whose lifetimes would exceed t (in the absence of such losses), without making any assumption about the form of the function $P(t)$. The observation for each item of a suitable initial event, marking the beginning of its lifetime, is presupposed.

For random samples of size N the product-limit (PL) estimate can be defined as follows: List and label the N observed lifetimes (whether to death or loss) in order of increasing magnitude, so that one has $0 \leq t_1' \leq t_2' \leq \cdots \leq t_N'$. Then $\hat{P}(t) = \prod_r [(N-r)/(N-r+1)]$, where r assumes those values for which $t_r' \leq t$ and for which t_r' measures the time to death. This estimate is the distribution, unrestricted as to form, which maximizes the likelihood of the observations.

Other estimates that are discussed are the actuarial estimates (which are also products, but with the number of factors usually reduced by grouping); and reduced-sample (RS) estimates, which require that losses not be accidental, so that the limits of observation (potential loss times) are known even for those items whose deaths are observed. When no losses occur at ages less than t, the estimate of $P(t)$ in all cases reduces to the usual binomial estimate, namely, the observed proportion of survivors.

Interval	Factor						
u_{j-1}, u_j	j	n_j	δ_j	λ_j	$\hat{p}_j$	$\hat{P}(u_j)$	$\hat{N}_{Ej}$
0 –1	1	100	3	0	97/100	.97	100
1 –1.7	2	97	5	20	92/97	.92	100
1.7–2	3	72	4	0	68/72	.87	88
2 –3	4	68	10	0	58/68	.74	83
3 –3.6	5	58	9	12	49/58	.63	80
3.6–4	6	37	6	0	31/37	.52	73
4 –5.0	7	31	15	16	16/31	.27	51

NONPARAMETRIC ESTIMATION FROM INCOMPLETE OBSERVATIONS

In the seminal paper by Kaplan and Meier (1958), survival estimates were presented in tabular form rather than as the Kaplan–Meier curve that is widely used today.

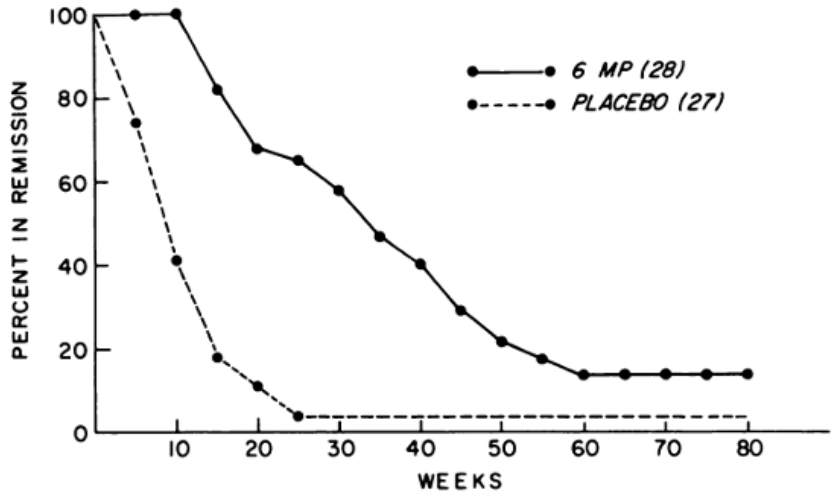


Fig. 2.—Effect of 6-MP and placebo maintenance therapy on the duration of complete remissions in Phase II.

The Effect of 6-Mercaptopurine on the Duration of Steroid-induced Remissions in Acute Leukemia: A Model for Evaluation of Other Potentially Useful Therapy

By 1963, Kaplan–Meier curves had become a standard way to present clinical trial results.

However, early plots did not include censoring marks or the number of patients at risk beneath the curve.

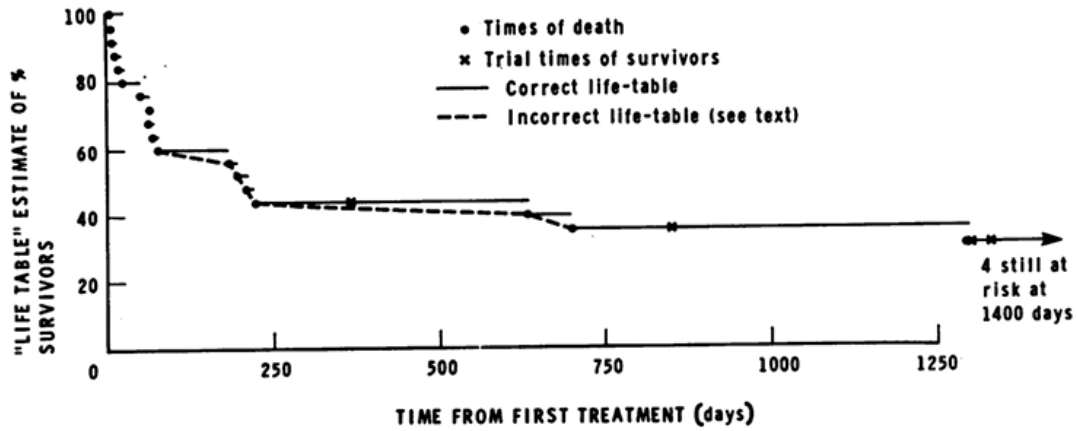


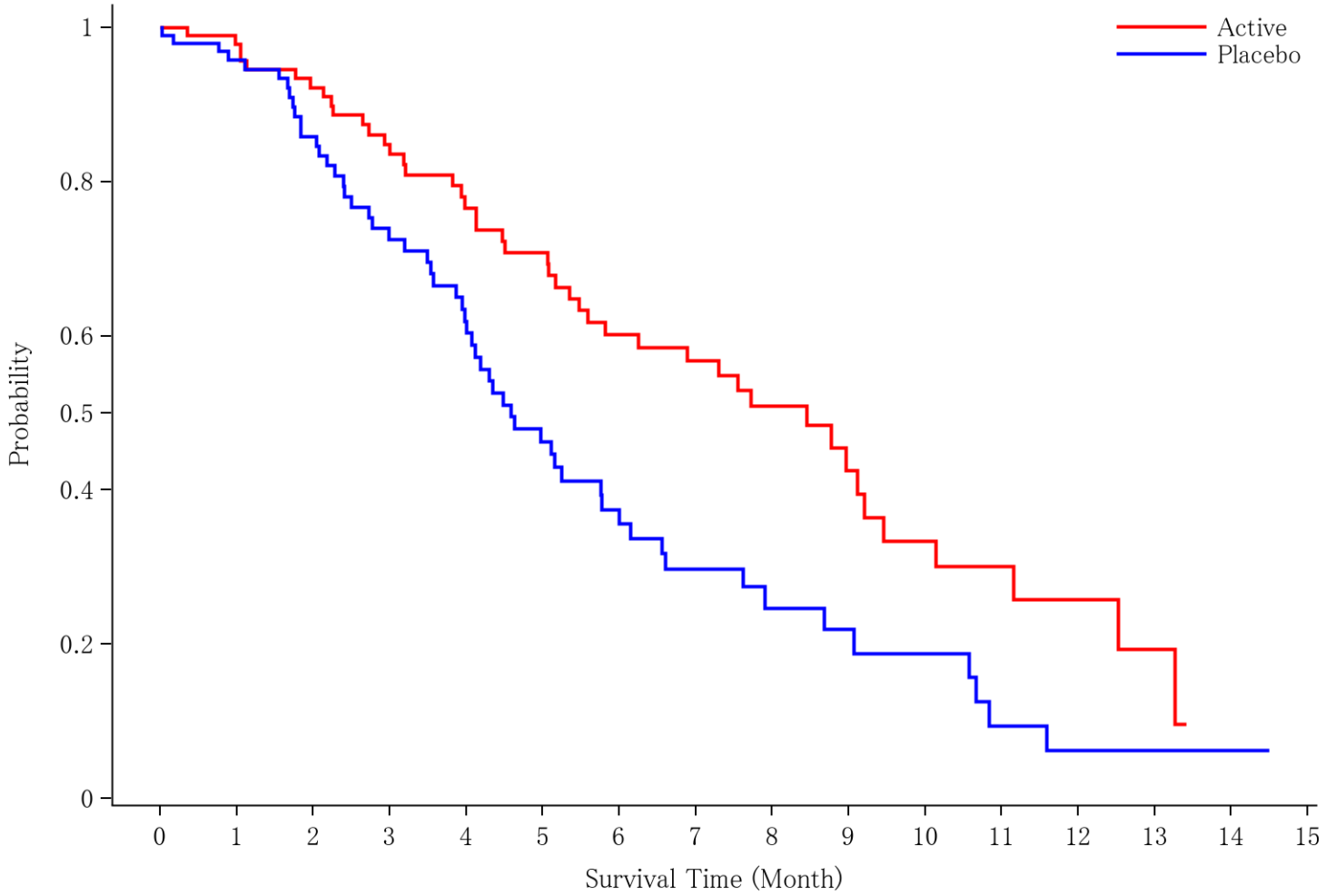
FIG. 5.—Life table derived from the hypothetical data used as a worked example in Appendix 3. The numbers of patients still alive and under observation at entry and every 6 months thereafter were: 25, 14, 11, 10, 8, 7, 7, 7, 4.

DESIGN AND ANALYSIS OF RANDOMIZED CLINICAL TRIALS REQUIRING PROLONGED OBSERVATION OF EACH PATIENT

To our knowledge, the earliest publication showing censoring marks on Kaplan–Meier curves could not be identified, possibly due to limitations in printing technology.

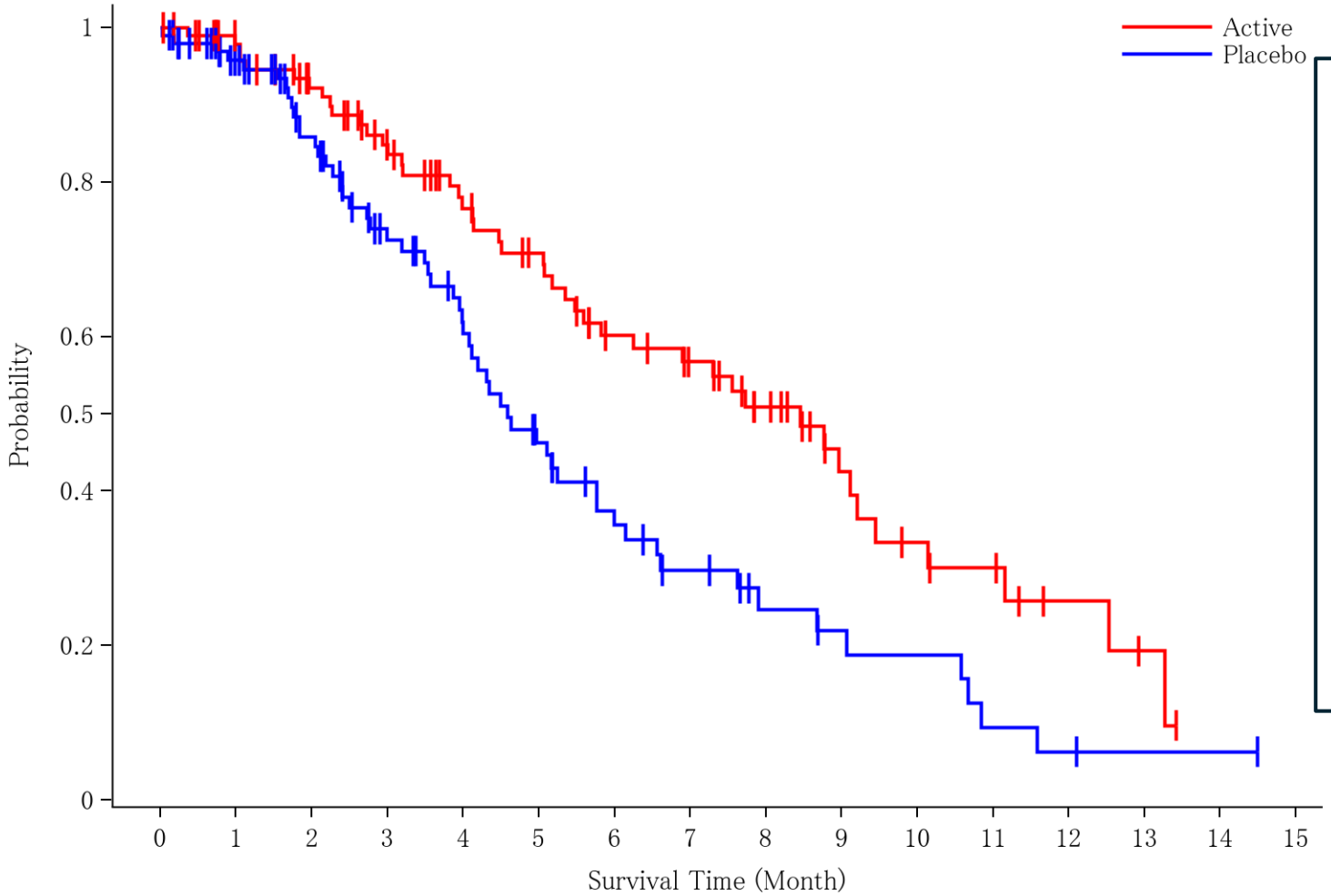
Such examples appear to have emerged in the 1970s or 1980s.

In Peto's statistical review (1976), censoring was represented in a style different from that used today, and the number of patients at risk was mentioned in a footnote.



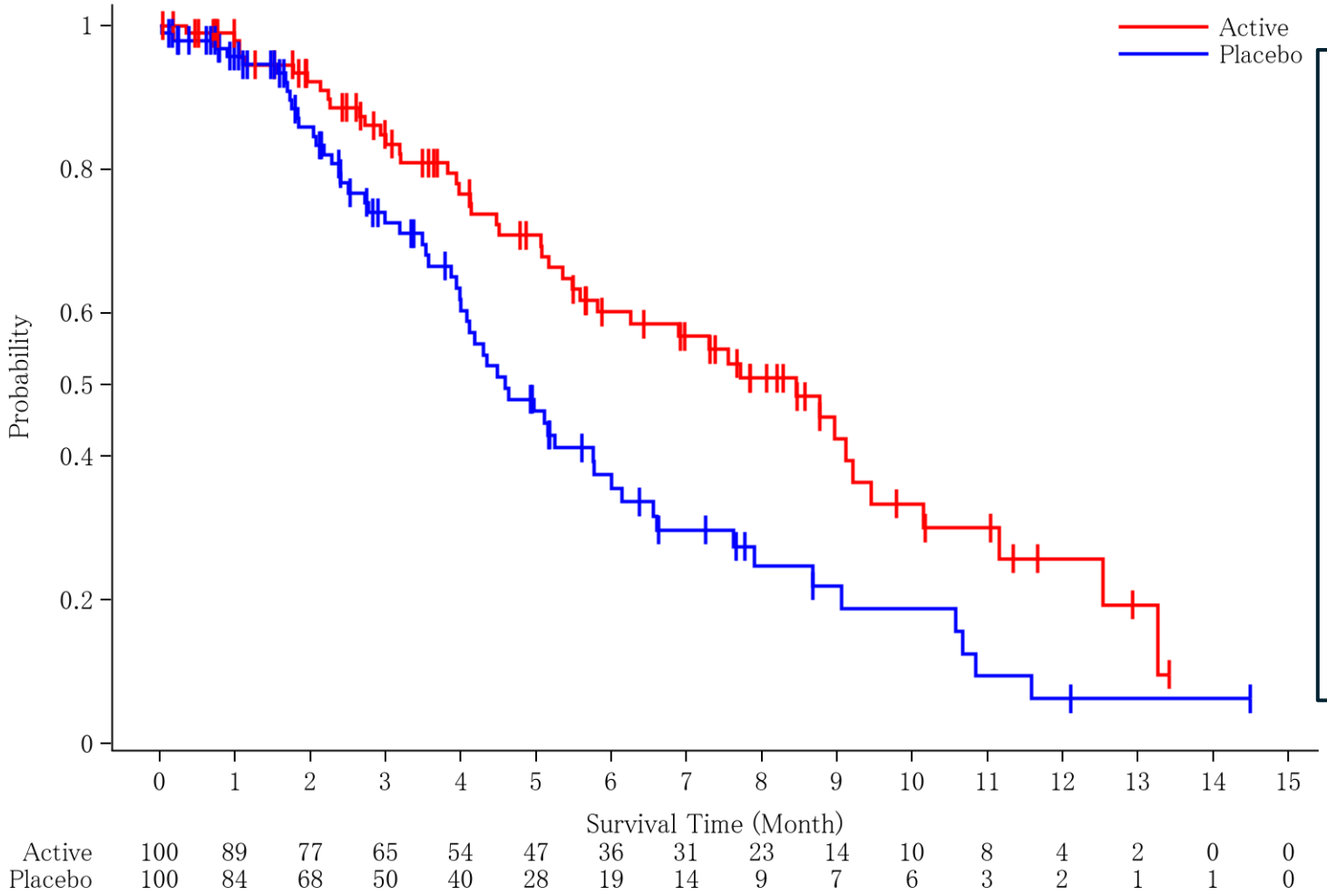
Early Kaplan–Meier figures appear to have consisted of plotted event points rather than continuous step curves.

Step curves later became the standard format, although censoring marks were generally omitted for many years.



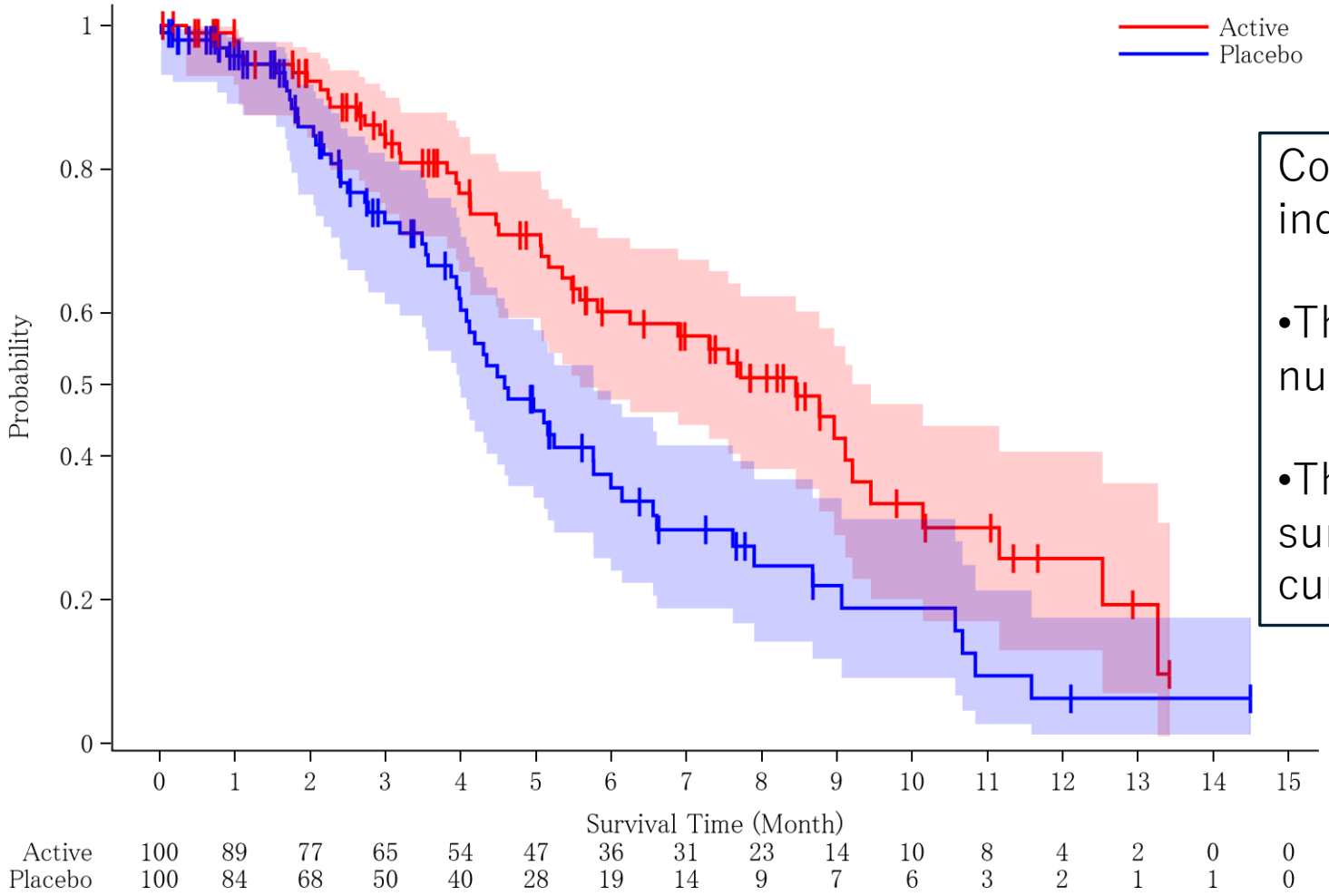
Censoring information is essential for the correct interpretation of Kaplan–Meier curves.

- It indicates where observations are no longer followed.
- It shows how much information supports the survival estimate over time.
- Without censoring marks, readers may overestimate the reliability of the tail of the curve.



The number at risk is essential for interpreting Kaplan–Meier curves.

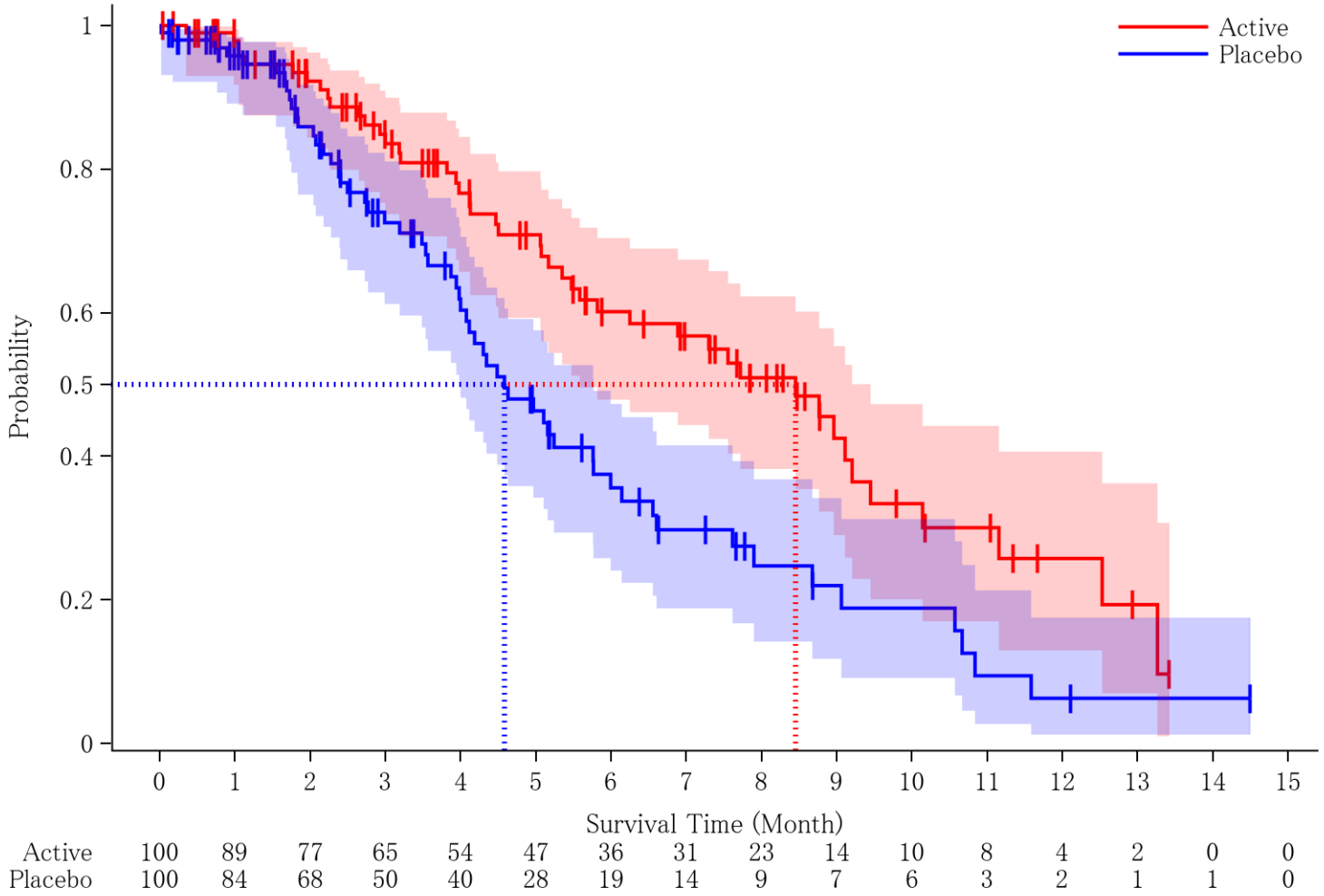
- It shows how many patients contribute to the survival estimate at each time point.
- It helps readers assess the precision and reliability of the survival estimates.
- A small number at risk indicates that the tail of the curve should be interpreted with caution.



Confidence interval bands provide a visual indication of statistical uncertainty.

•The confidence intervals generally widen as the number at risk decreases.

•This helps visualize the increasing uncertainty of survival estimates, particularly in the tail of the curve.



Modern Kaplan–Meier plots often incorporate additional visual elements to improve interpretation.

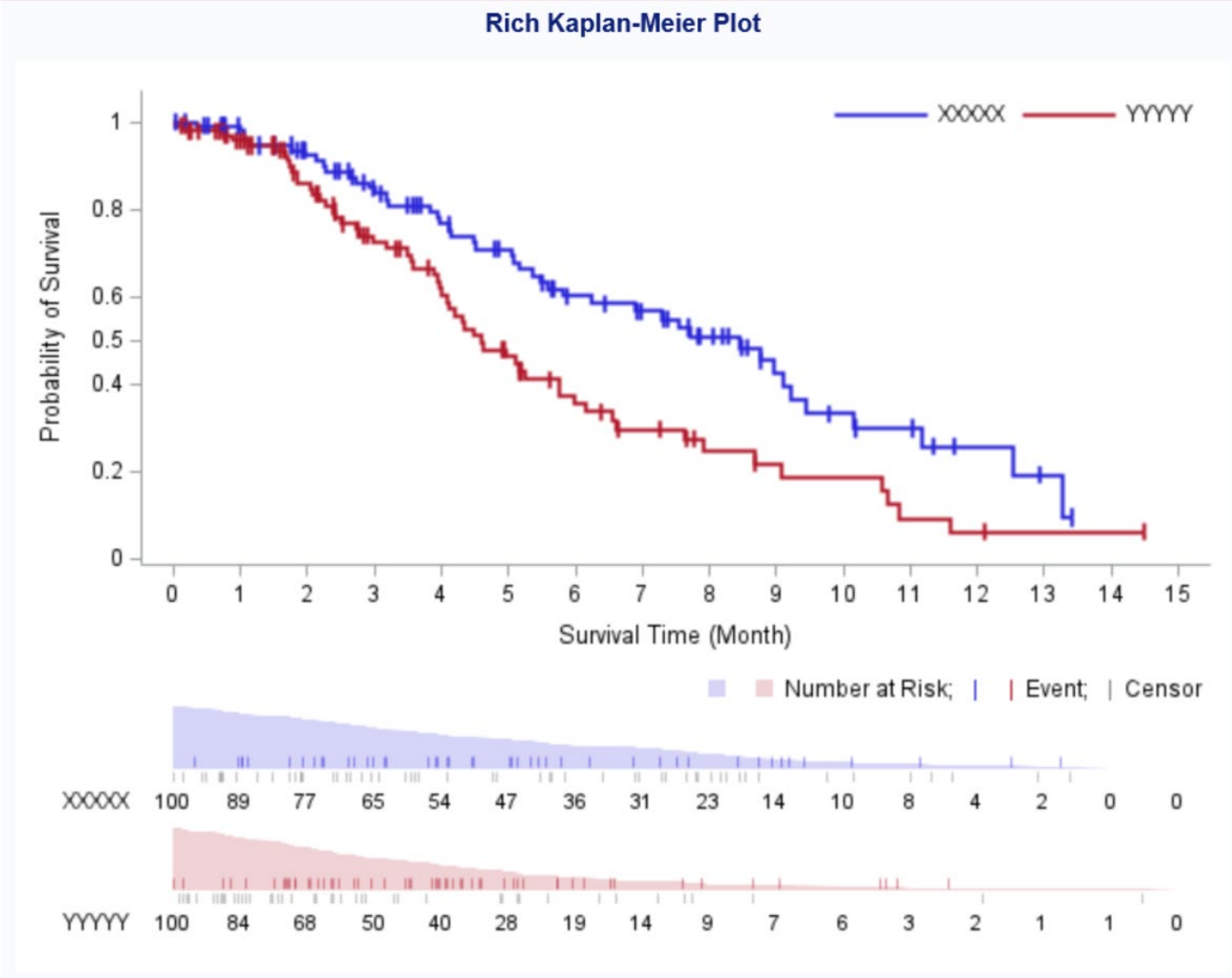
- Droplines can be added to highlight the median survival time.
- Hazard ratios and log-rank test results are often displayed as annotations.
- These enhancements provide more information without requiring separate tables or figures.

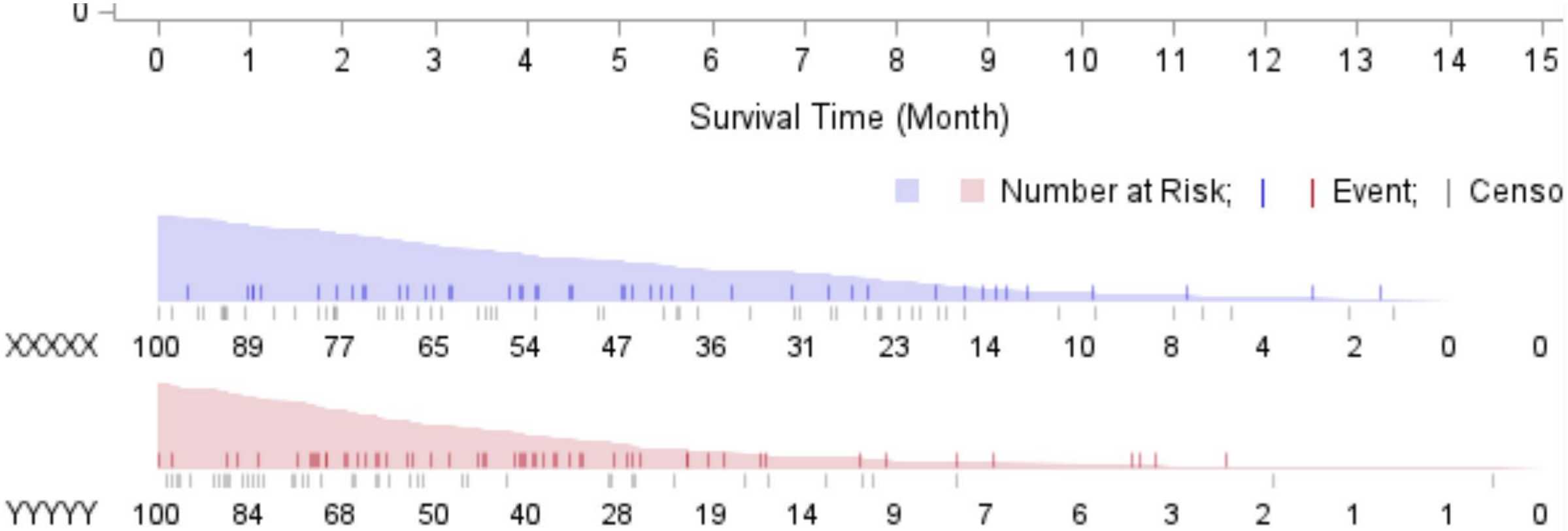
Limitations of Conventional Kaplan–Meier Plots

- Readers must mentally integrate information from both the survival curve and the number-at-risk table.
- The timing and accumulation of censoring events are difficult to recognize.
- Confidence interval bands alone do not adequately convey the increasing uncertainty in the tail of the curve.



To overcome these challenges, I developed the Rich Kaplan–Meier Plot.





Visualizing the Number at Risk

- The number at risk is normalized within each group (100 at baseline and 0 at the final observation).
- Changes in the number at risk are represented as a continuous volume along the time axis.
- This provides an intuitive visual impression of the diminishing information available over time.



rich_kaplan_meier_plot

This package provides a Rich Kaplan-Meier plot for SAS, enhancing standard survival curves with censoring marks, event marks, standardized number-at-risk bands, and risk tables. It helps visualize survival probability and risk-set dynamics across up to four groups in an intuitive, presentation-ready format.

https://github.com/PharmaForest/rich_kaplan_meier_plot

Making Rich Kaplan–Meier Plots Accessible
•I have released Rich Kaplan–Meier Plot as a SAS package.
•Any SAS user can easily create Rich Kaplan–Meier plots with minimal coding.

`%rich_kaplan_meier_plot()` macro

Purpose: This macro generates a rich Kaplan-Meier survival plot using PROC LIFETEST in SAS.

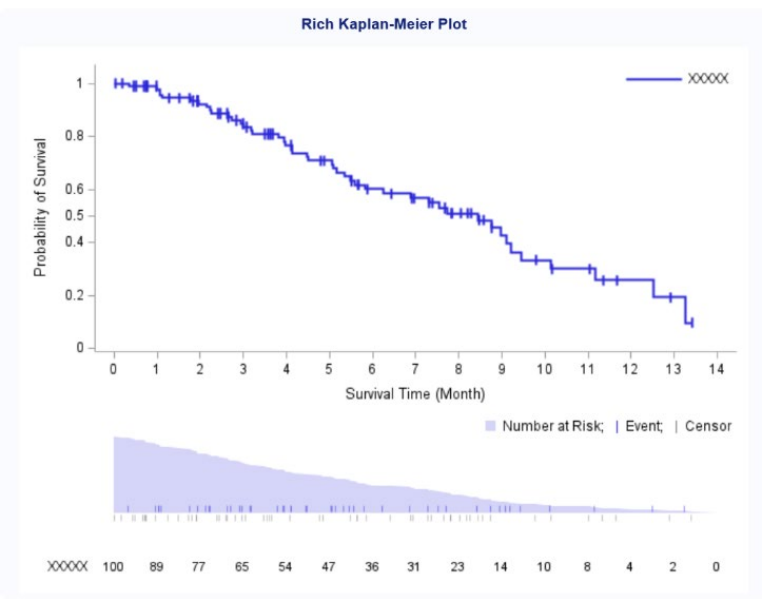
```
It produces survival curves by group, displays censoring marks, event marks,
number-at-risk bands, and number-at-risk tables in a split-panel layout.
```

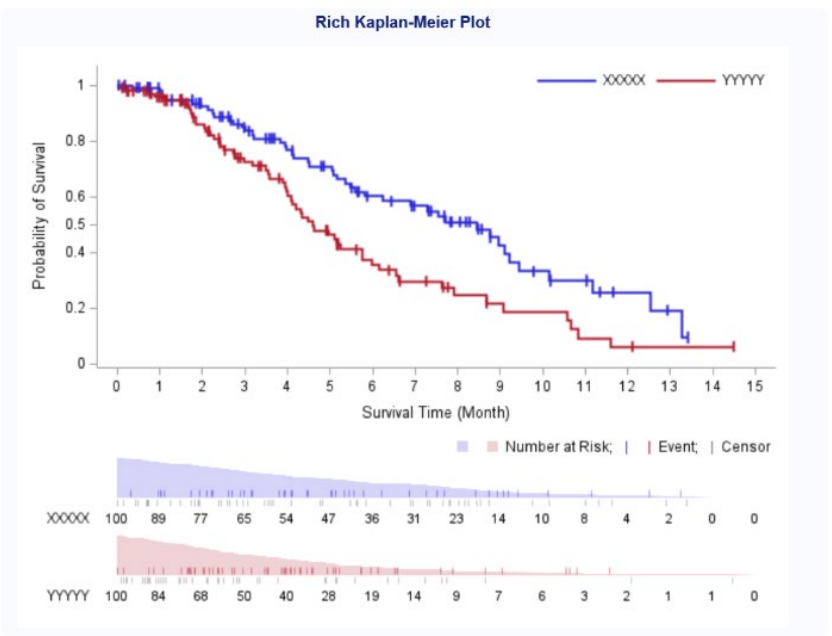
Parameters:

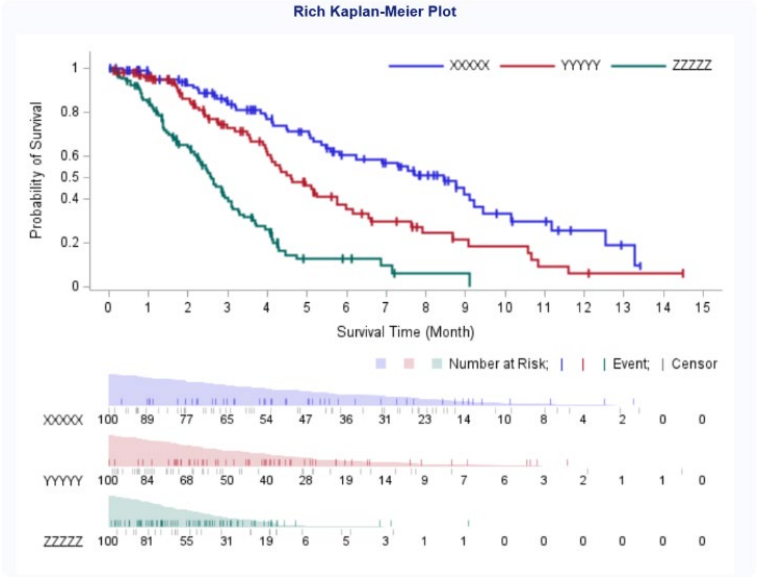
```
data=           Input dataset name (default: dummy_adtte)
wh=             WHERE condition to subset data (optional)
groupn=         Numeric group variable used for stratification (e.g., TRTPN)
groupc=         Character group label variable (e.g., TRTP)
idvar=          Subject identifier variable used for tooltips (e.g., USUBJID)
Time_var=       Time-to-event variable (e.g., AVAL)
Censor_var=     Censoring indicator variable (e.g., CNSR)
Censor_val=     Value indicating censored observations (e.g., 1)
Title=          Plot title (default: "Kaplan-Meier Plot")
XLABEL=         Label for the X-axis (e.g., "Survival Time (Month)")
YLABEL=         Label for the Y-axis (e.g., "Probability of Survival")
AxisValues=     Tick marks for the X-axis (e.g., "0 1 2 3 4 5")
Generate_Code=  Option to output MFILE-generated SAS code (Y/N)
```

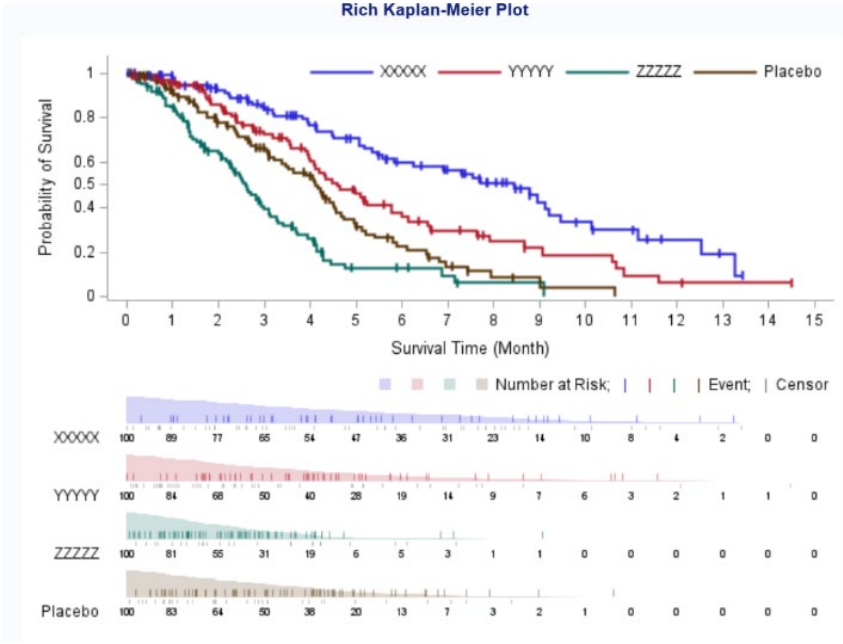
Example usage:

```
%rich_kaplan_meier_plot(
    data = dummy_adtte,
    wh = %nrbquote(TRTPN in (1: 2)),
    groupn = TRTPN,
    groupc = TRTP,
    idvar = USUBJID,
    Time_var = AVAL,
    Censor_var = CNSR,
    Censor_val = 1,
    Title = %nrbquote(Rich Kaplan-Meier Plot),
    XLABEL = %nrbquote(Survival Time (Month)),
    YLABEL = %nrbquote(Probability of Survival),
    AxisValues = %nrbquote(0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15),
    Generate_Code = Y
);
```



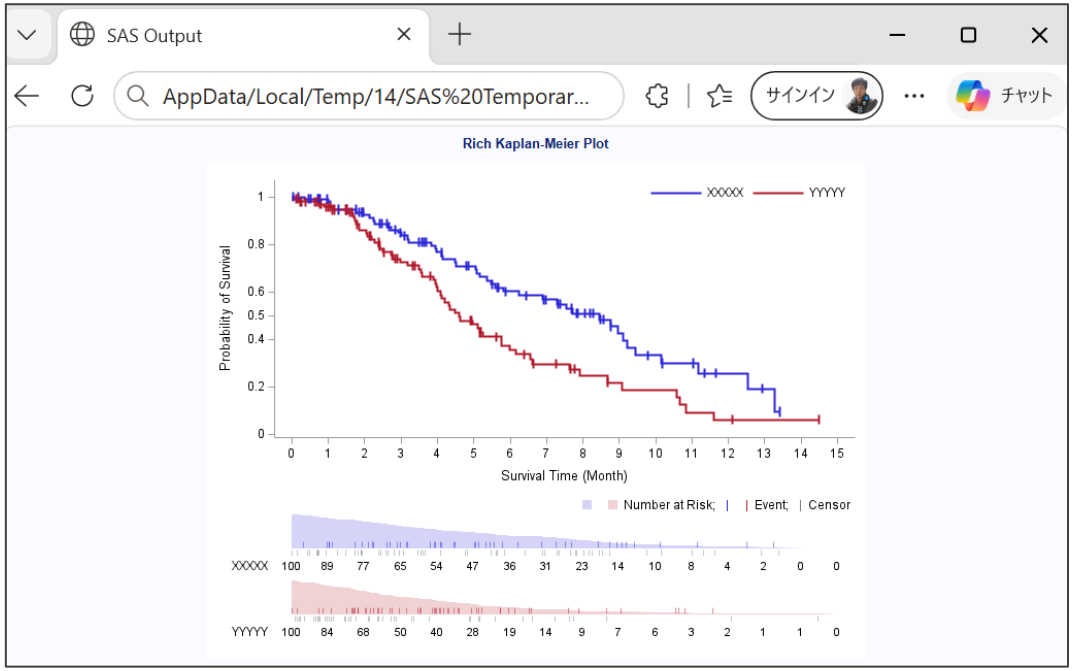






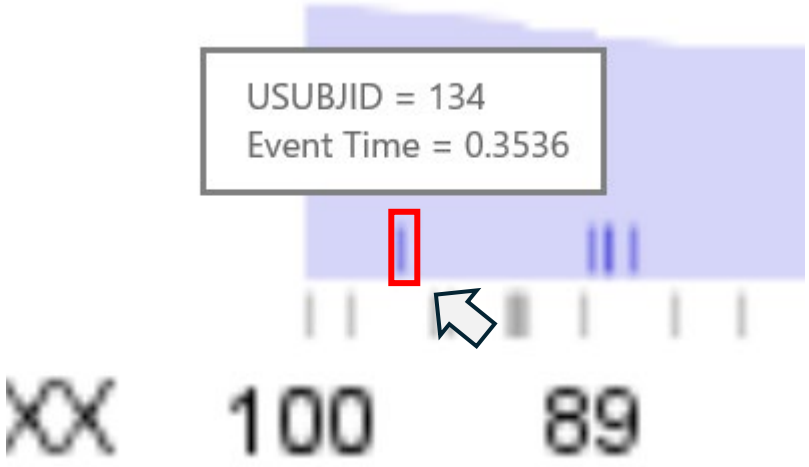
Flexible Support for Multiple Groups

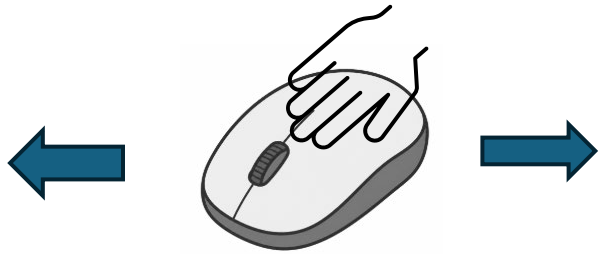
- The `rich_kaplan_meier_plot` package supports plots with one to four groups.
- In our experience, one to three groups provide the best balance between information density and readability.
- Plots with four or more groups may become visually cluttered and more difficult to interpret.

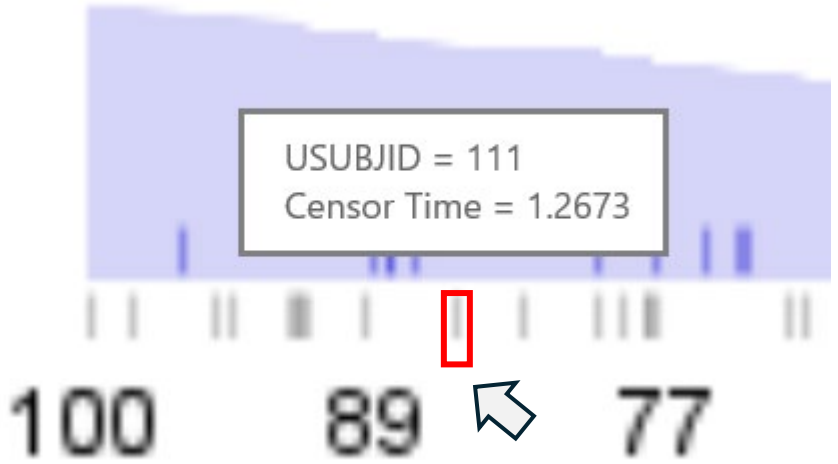


Interactive Rug Plot

- The rug plot makes the distribution of events and censored observations easy to recognize.
- In the generated HTML output, hovering over a rug mark displays detailed information for the corresponding subject.
- The popup shows whether the observation is an event or censoring, together with the corresponding time point.





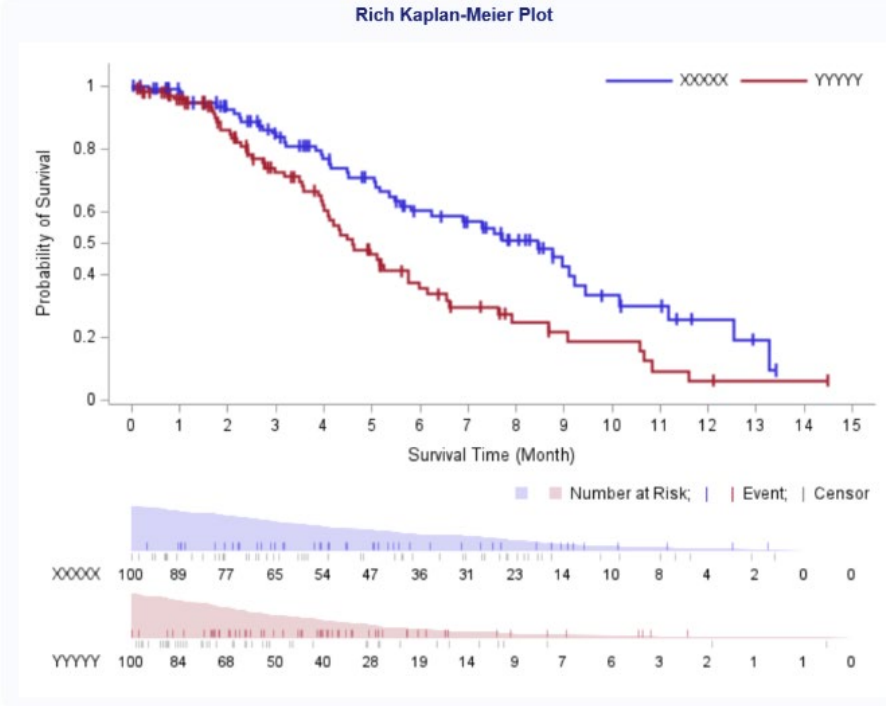


When Generate_Code=Y, the SAS code that actually generates the plot can be exported as plain code.

rich_kaplan_meier_plot19.txt - メモ帳

ファイル(F) 編集(E) 書式(O) 表示(V) ヘルプ(H)

```
run;
proc sort data=_dummy_adtte(keep=TRTPN TRTP) out=group_fmt nodupkey;
by TRTPN TRTP;
run;
data group_fmt;
set group_fmt;
FMTNAME = "$RKM_GR";
START = cats(TRTPN);
LABEL = TRTP;
call symputx(cats("label",TRTPN ),LABEL);
run;
proc format cntlin=group_fmt;
run;
ods graphics on;
ods noresults;
ods select none;
ods output Survivalplot=SurvivalPlotData;
ods output ProductLimitEstimates =ProductLimitEstimates;
proc lifetest data=_dummy_adtte plots=survival(atrisk=0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 cl);
time AVAL * CNSR(1);
strata TRTPN ;
id USUBJID;
run;
ods results;
ods select all;
proc sql noprint;
select max(StratumNum) into :max_StratumNum from SurvivalPlotData;
quit;
proc sort data=SurvivalPlotData(keep = Stratum) out=Stratum nodupkey;
by Stratum;
run;
proc sort data=SurvivalPlotData(keep = tAtRisk) out=tAtRisk nodupkey;
where ^missing(tAtRisk);
by tAtRisk;
run;
data atrisk;
set Stratum;
if _N_=1 then do;
declare hash h1(dataset:"SurvivalPlotData(keep=Stratum tAtRisk)");
h1.definekey("Stratum","tAtRisk");
h1.definedone();
end;
```



Fully Transparent Code Generation

- Complex visualization tools often encourage users to understand how the plots are generated.
- Users may also wish to customize the implementation to suit their own preferences.
- After execution, the package automatically outputs the complete plotting code in plain SAS, making every step transparent and fully customizable.

README.md

PharmaForest (we have already 45 packages 🎉 🎉)

A collaborative repository of SAS packages for pharmaceutical industry, powered by PHUSE Data Visualization & Open-Source Technology(DVOST) Working Group.



For more details about PharmaForest and the DVOST WG activity, please visit the PHUSE page:
PharmaForest: A Collaborative Repository of SAS Packages for Pharmaceutical Industry

https://github.com/PharmaForest

Part of the PharmaForest Ecosystem

- The `rich_kaplan_meier_plot` package is part of the PharmaForest ecosystem.
- PharmaForest is an open-source collection of SAS packages designed for pharmaceutical data analysis.
- It is built on the SAS Packages Framework, which simplifies package distribution, installation, and maintenance.

https://pharmaforest.github.io/



Wellcome Open Research

Search

BROWSE GATEWAYS & COLLECTIONS HOW TO PUBLISH ABOUT BLOG

74484 Views | 6124 Downloads | 0 Citations | CITE | DOWNLOAD

Home › Articles › Raincloud plots: a multi-platform tool for robust data visualization

SOFTWARE TOOL ARTICLE

REVISED Raincloud plots: a multi-platform tool for robust data visualization

[version 2; peer review: 2 approved]

Micah Allen, Davide Poggiali, Kirstie Whitaker, Tom Rhys Marshall, Jordy van Langen, Rogier A. Kievit

ARTICLE AUTHORS METRICS

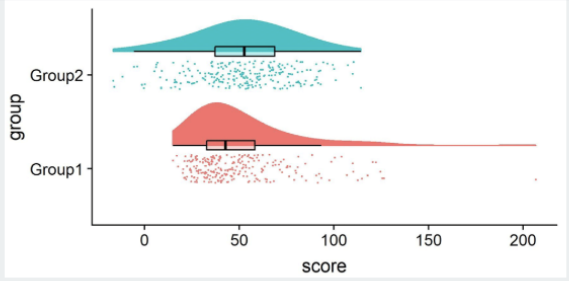
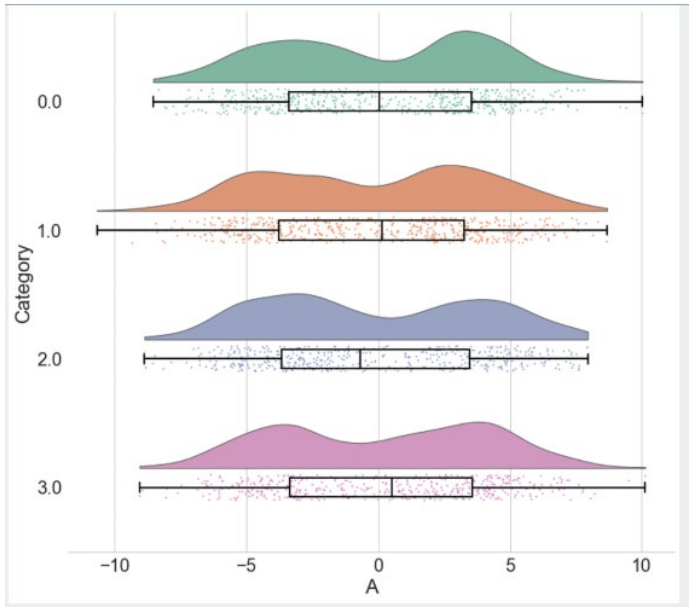


Figure 3. Example Raincloud plot.

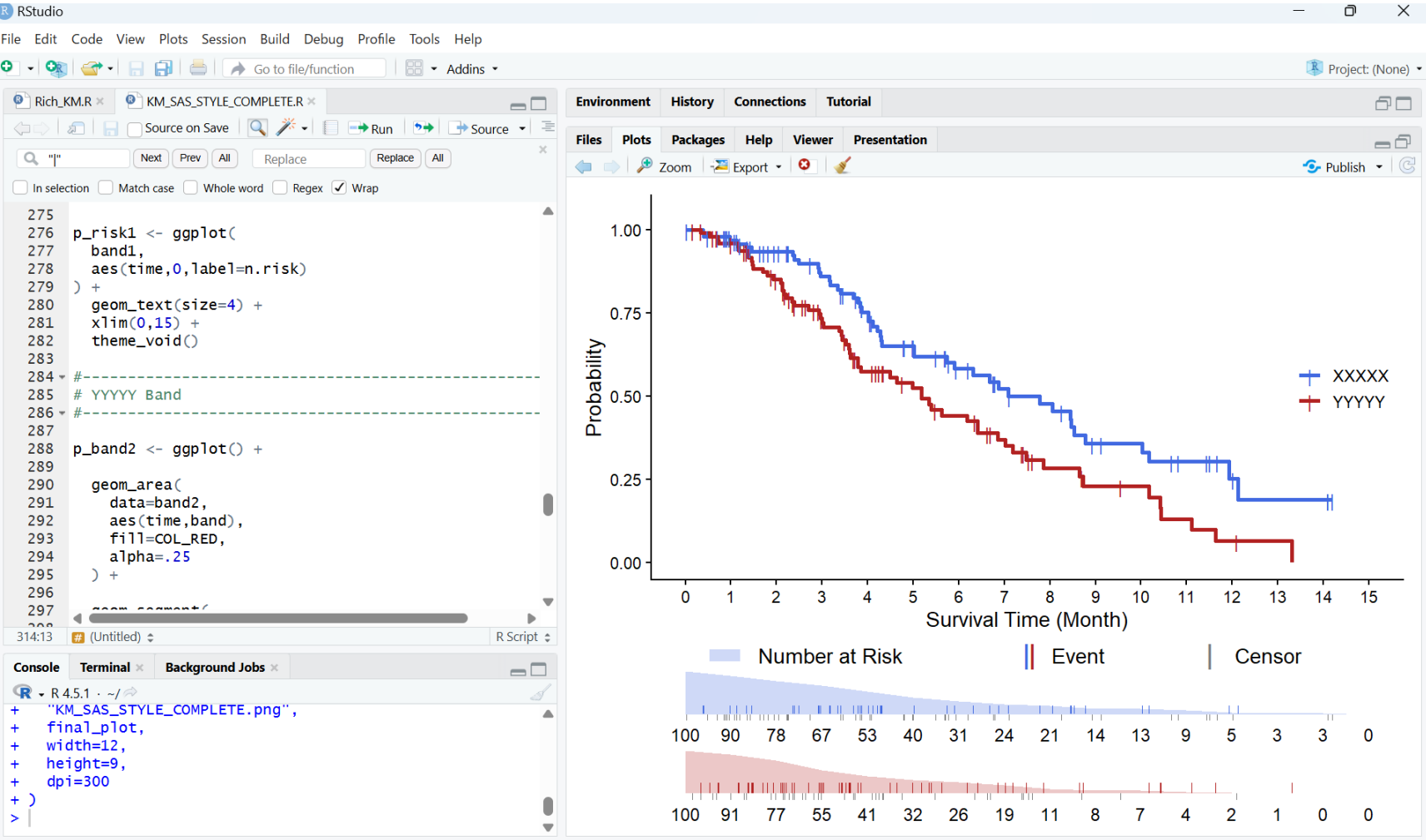
The raincloud plot combines an illustration of data distribution (the 'cloud'), with jittered raw data (the 'rain'). This can further be supplemented by adding boxplots or other standard measures of central tendency and error. See figure3.Rmd for code to generate this figure.



https://wellcomeopenresearch.org/articles/4-63

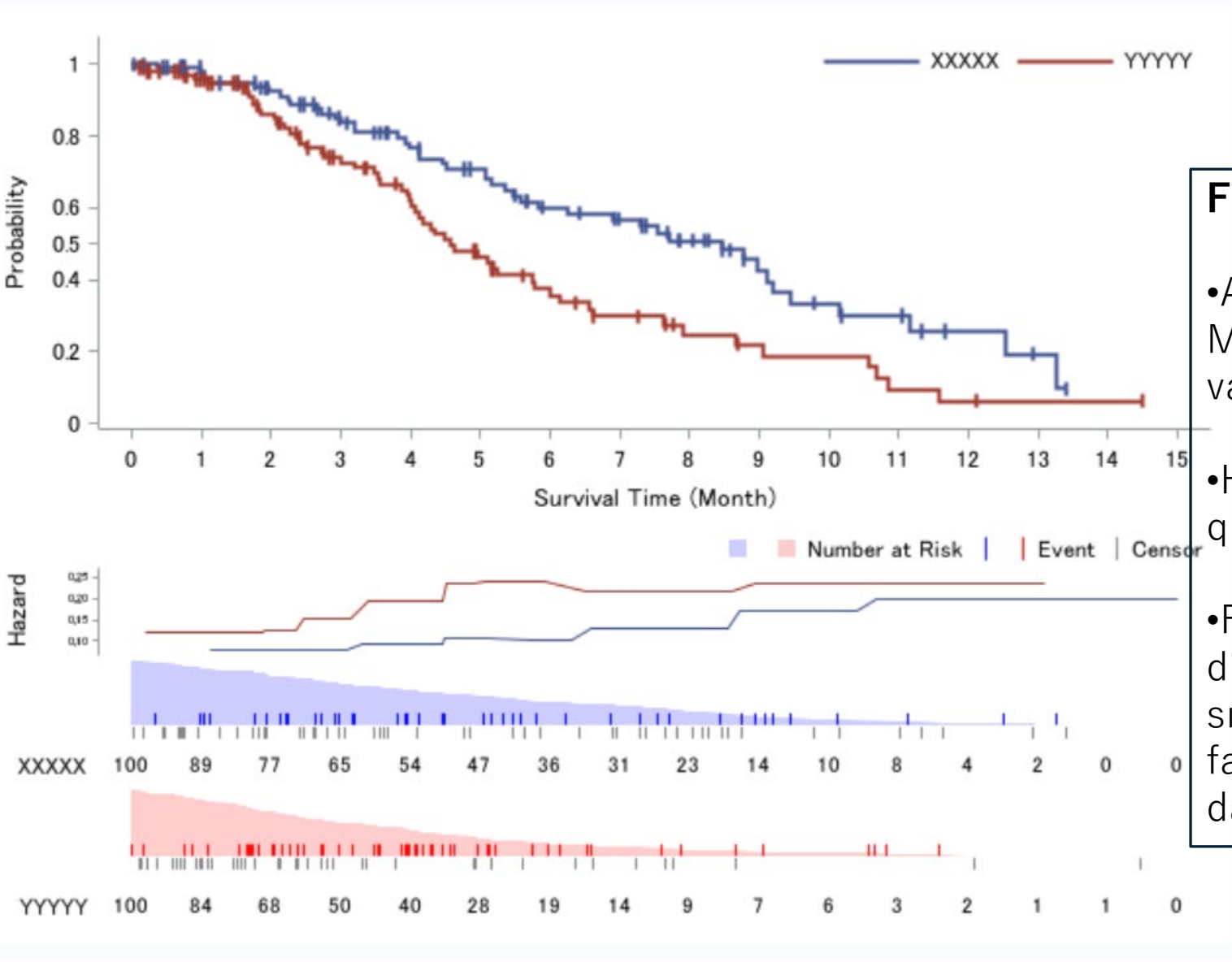
Inspired by the Raincloud Plot

- My commitment to open-source development was inspired by the pioneering work of Mika Allen and colleagues on the Raincloud Plot.
- They published their work in an open-access journal and provided implementations in multiple programming languages.
- This made the method widely accessible, leading to rapid adoption across the scientific community.



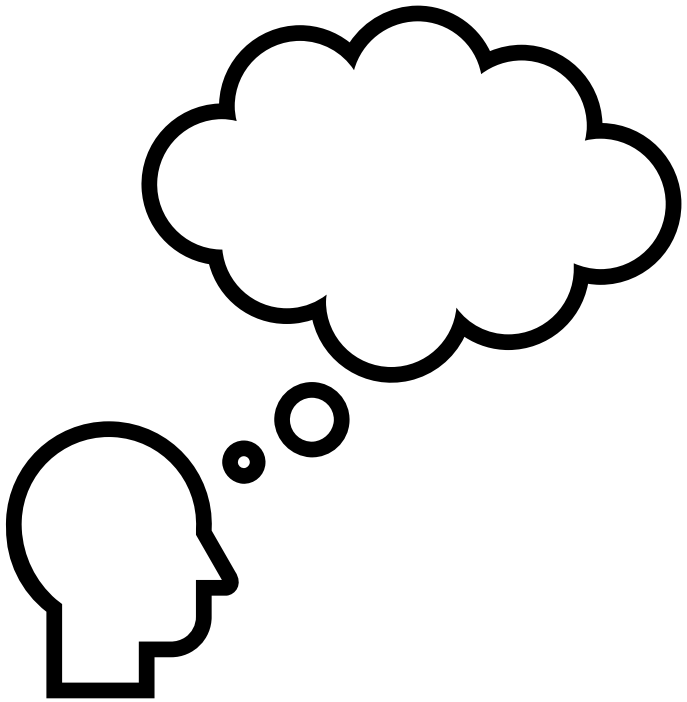
Future Directions , in R.

- Following the philosophy of Mika Allen and colleagues, we are extending Rich Kaplan–Meier Plot beyond SAS.
- We have demonstrated that the plot can also be implemented in R.
- Future plans include developing an R package and an interactive Shiny application to make the method accessible to an even broader community.



Future Research

- A future extension of the Rich Kaplan–Meier Plot is the visualization of the time-varying hazard.
- However, several methodological questions remain.
- For example, what is the analytical value of displaying hazards, and what degree of smoothing best balances visual clarity with faithful representation of the underlying data?



Future Possibilities

- Different symbols could be used to distinguish different types of censoring (e.g., administrative vs. dropout).
- Time-varying hazard ratios could be visualized alongside the survival curves.
- Visualize event intensity as a density layer rather than individual rug marks.
- Display cumulative differences in restricted mean survival time (RMST).
- Extend the visualization to competing-risk analyses.

Kaplan–Meier curve is not a finished visualization.

- The Kaplan–Meier plot is still evolving rather than a finished visualization.
- There is still considerable room for improving how survival analyses are communicated.
- Through an open-source ecosystem, we hope the community can continue to refine and extend this visualization together.
- We hope Rich Kaplan–Meier Plot will become one small step toward that future.

Thank you

The Kaplan–Meier plot is still evolving.
Let's continue improving survival visualization together through open science.